Network monitoring

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Abstract: Networks

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Introduction

Here, we used simulations to test the efficiency of monitoring species interactions and species richness within BONs, testing several methods for design and optimization.

- Context
    - Biodiversity monitoring should encompass all facets of biodiversity
    - Species interactions are rarely considered in comparison to species richness metrics, in part due to their inherent sampling difficulty
    - Yet, interactions represent an biodiversity component which should be considered when implementing monitoring networks
    - As interactions are inherently challenging to sample, developing methods for optimizing strategies is even more relevant.
- Key questions
    - How should we implement monitoring programs *across space* to efficiently capture network composition and changes as essential components of biodiversity?
    - Can we monitor species interactions as efficiently as species ranges? What would be realistic expectations for monitoring species interactions?

Methods

We explored network sampling using simulations to generate networks, species ranges, and biodiversity observations networks (BONs). We followed the approach introduced by Catchen et al. (2023) to simulate interaction realization and detection, implemented in SpeciesInteractionSamplers.jl.

First, we simulate networks with variation across space. To do so, we generate autocorrelated ranges representing the distribution of species across a simulated landscape and generate a metaweb using the niche model to define the feasible interactions between the species. Combining the species ranges and the metaweb information, we obtain the list of **possible interactions** in our landscape and inside every pixel based on the co-occurrence of species with feasible interactions. Next, we incorporate a consideration for species abundances, which influence both interaction realization and detection. For example, rare and low-abundance species are less likely to interact in a location because they are less likely to encounter one another, especially compared to abundant species, leading to neutrally forbidden links (Canard et al. 2012; Catchen et al. 2023). We generate an abundance profile for our set of species (based on a Normalized Log Normal distribution), then draw **realized interactions** at each location given the abundance of species. Finally, we define the set of **detected interactions** by conditioning the detection of realized interactions on the abundance distribution (even if an interaction is realized in a location, we need to be there at that moment to detect, which is again less likely for low probability events; Catchen et al. (2023)).

Second, we design Biodiversity Observation Networks (BONs) across the simulated landscape and measure their efficiency at monitoring the generated networks when increasing the number of sites in the BON. We test different algorithms to spread monitoring sites across the landscape and optimize given different information such as species ranges, species richness or presence of realized interactions.

Across our simulations, we used landscapes of 100 x 100 pixels, 75 species, and a connectance of 0.2 to generate metawebs using the niche model.

Evaluating metaweb sampling

First, we investigated the efficiency of sampling networks across space to document a metaweb of interactions, comparing it with the efficiency to document species richness. We generated spatially balanced Biodiversity Observation Networks (BONs) across our landscape, defining sites where to sample species and interactions. We tested all number of sites between 1 and 100 and used the BalancedAcceptance method, which aims to distribute sampling evenly across space (for a given number of sampling sites). We then evaluated the sampling efficiency of each BON in terms of overall species richness and compared to the generated metaweb for each interaction type. For species richness, we assumed we sampled all species present in a sampled location, combined this information across all sampled sites to determine the number of species encountered through sampling, and compared the result with the total species richness in the landscape (75 species). For interactions, we aggregated each interaction type separately (possible, realized and detected) for a given BON and evaluated the proportion of the metaweb recovered through sampling (and therefore for a given number of site).

We ran simulations with 100 fully-independent replicates (different metaweb, species ranges, abundance distribution, realized interactions) for each number of site in a BON and for the two interaction references, yielding 10,000 simulations in total. Prior exploratory simulations showed the number of sites in the BON to be the main element affecting the proportion of sampled species or interactions compared to other model parameters such as metaweb connectance and landscape autocorrelation.

Focal sampling on single-species & optimization

In complement to the metaweb sampling approach, we also investigated the sampling efficiency on a single-species basis. We simulated a focal sampling where an observer would target one species and aim to retrieve all of its interactions (here known from the metaweb), but might have different information to optimize its sampling. Observation network design and efficiency needs to be evaluated beyond spatial balance, and also regarding their efficiency at meeting biodiversity monitoring objectives (Norman and Poisot 2025). Thus, the idea of our focal sampling exploration is to represent a simplified case with a clear objective (retrieving a species' interactions) and where we do have prior knowledge about the system, either through empirical data or model predictions, which can be used to optimize sampling.

Optimizing focal sampling relies on two elements: the sampler algorithm used and the information provided for optimization. Species range represents an attainable level of information an interested observer could have, for instance based on expert knowledge or species distribution models. Therefore, we assumed the observer knew the species range across the landscape (presence or absence). We tested three sampling algorithms for comparison: simple random sampling (to represent uninformed sampling), weighted balanced acceptance sampling (aiming to balance sites across space while weighting favorably for species ranges) and uncertainty sampling (which targets locations with a higher value in the layer provided). Next, we tested four information layers for optimization: one with the focal species range (our attainable information knowledge for the observer), one with the probabilistic range (for a less informed case, similar to the output of a species distribution model), one with the location of realized interactions for the focal species (to represent a best-case scenario) and one with species richness across the landscape (to represent a holistic level of information for the system).

Using the approach described in the previous section, we generated a single* set of metaweb, species ranges, and locations for realized interactions (*counter-verified with 10 separate sets, which yielded conceptually similar results). We selected the species with the highest degree (number of interactions) in the metaweb as the focal species for the simulation. We then generated BONs of 1 to 500 sites (at interval increments of 5 sites) using the three possible algorithms and optimization layers, with 100 replicates for every number of sites. For each BON generated, we extracted the number of unique realized interactions sampled across all sites (essentially the monitored degree for the focal species)

Efficiency

Next, we aimed to verify whether our results were constant when replicating simulations with different simulated metawebs and species ranges. We ran 100 independent simulations, each with 50 replicates for every number of sites (once again from 1 to 500 sites at interval increments of 5 sites). However, doing so raised the issue of how to describe and compare the efficiency of the simulated monitoring across samplers, optimization layers or focal species degree, given how the generated metawebs and landscapes might vary between independent simulations, and especially how to describe the relationship between the number of monitoring sites and the proportion of monitored interactions.

Since the exact proportion of monitored interactions is known in our simulations, we used the following single-parameter equation which generates similar efficiency curves:

$$p = \frac{x}{a + x} \tag{1}$$

where p is the proportion of monitored interactions, x is the number of sites in the BON and a is the efficiency parameter we use for comparison. Here, a is the single parameter describing the shape of the curve, and can be compared across samplers, optimization layers or focal species degree. The lower this parameter, the faster we reach a high proportion of monitored interactions.

For each independent simulation, sampler and optimization layer option, we performed a grid search across possible values of a to identify the curve yielding the best fit with the simulation's result. Upon exploratory analysis, we observed that searching for values of a on an exponential scale allowed for more gradual control over the curve shape, and therefore we searched we searched directly for values of e^a (effectively replacing a by e^a in Equation 1). We performed a grid search of 10,000 evenly spaced values between −5 and 15 (limits chosen to go beyond the fitted values), and selected the value of a yielding the curve minimizing the sum of squared errors to the values from the simulation.

To compare the efficiency curves within a given simulation (i.e., between samplers and optimization layers for a given metaweb and set of species ranges), we used the definite integral of Equation 1 between 0 and k, the maximum number of sampling sites in the landscape (10,000), which takes the form:

$$\int_0^k \frac{x}{a + x}\,\mathrm{d}x = a\ln(a) - a\ln(k + a) + k \tag{2}$$

Using Equation 2, we can compute the difference in efficiency between two curves as the difference of integrals:

$$d_{1,2} = (a_1 \ln(a_1) - a_1 \ln(k + a_1) + k) - (a_2 \ln(a_2) - a_2 \ln(k + a_2) + k) \tag{3}$$

This difference measure allows to identify which samplers or optimization layer is more efficient, quickly reaching a high proportion of monitored interactions (positive values indicate that the first curve is more efficient than the second), while also quantifying the magnitude of this difference in a comparable way between possible comparison pairs. We compared all samplers and optimization layers individually within each independent simulation, given that the monitoring context is different in each one (different metawebs and species ranges).

Results

Lower sampling expectations for species interactions than species richness

Our results for metaweb sampling highlight a marked difference between the proportions of sampled species and interactions (Figure 1). Sampled species saturated very quickly, with all species sampled with less than 25 sites in a BON across the landscape. Possible interactions, which depend on co-occurrence of species with a feasible interaction in the metaweb, also saturated quickly, reaching over 90% of interactions in the metaweb with 25 sites and increasing asymptotically afterwards. In contrast, realized and detected interactions, which account for the impact of species abundances, reach much lower proportions, both under 25% of the metaweb after 100 sites. While the proportion would keep increasing with a higher number of sites, the speed at which interactions accumulate is notably lower than for species richness and possible interactions, highlighting the need for a much higher expected sampling effort.

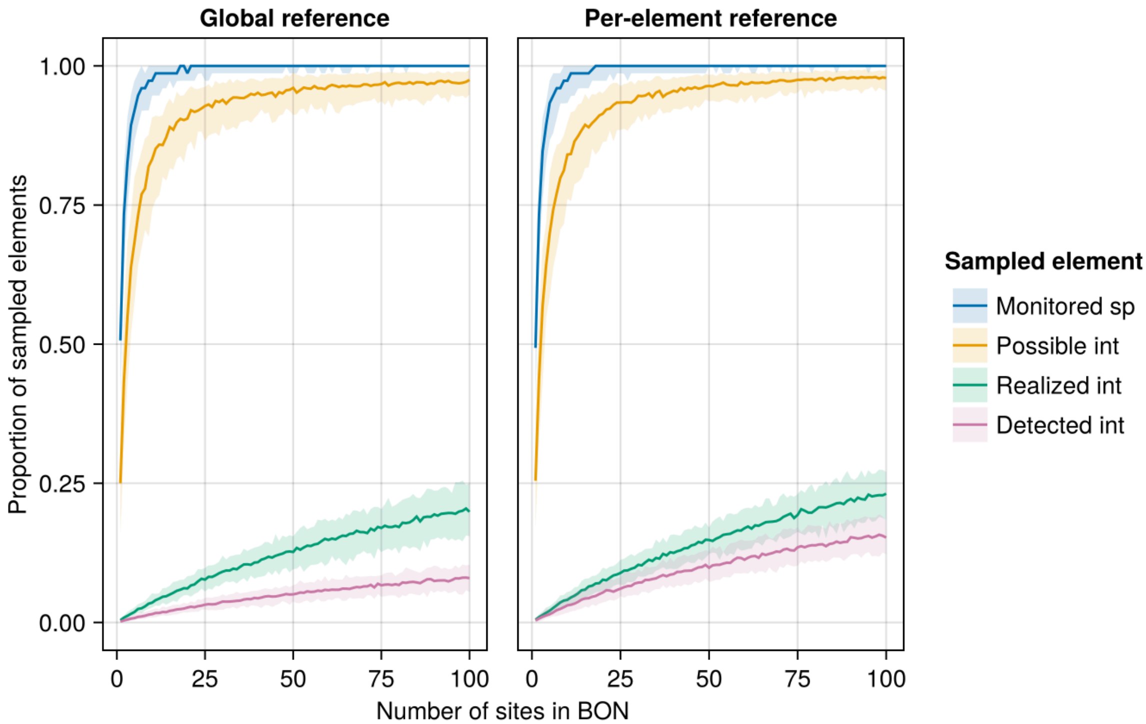

Figure 1: Proportion of sampled species and interactions for a given number of sampling sites in a Biodiversity Observation Network (BON) across the simulated landscape. The central line indicates indicates the median proportion for a given number of sites while the shaded bands delimit the 5th and 95th percentile across replicates. The global reference (left) uses total species richness and the total number of interactions in the metaweb as the maximum reference to calculate the proportion while the per-element reference (right) uses the maximum number of an interaction actually present in the simulated landscape.

Differences between samplers and optimization layers

Our focal sampling experiment showed the Uncertainty Sampling algorithm to be most efficient at optimizing BON-design based on a known species range and while aiming to maximize the sampling of interactions of a single focal species (Figure 2). Simple Random sampling, as a proxy for uninformed sampling, performed the worst among the algorithms tested. Weighted Balanced Acceptance performed notably less than Uncertainty Sampling at maximizing the efficiency for the focal species, although it maintained a better spatial balance across the landscape.

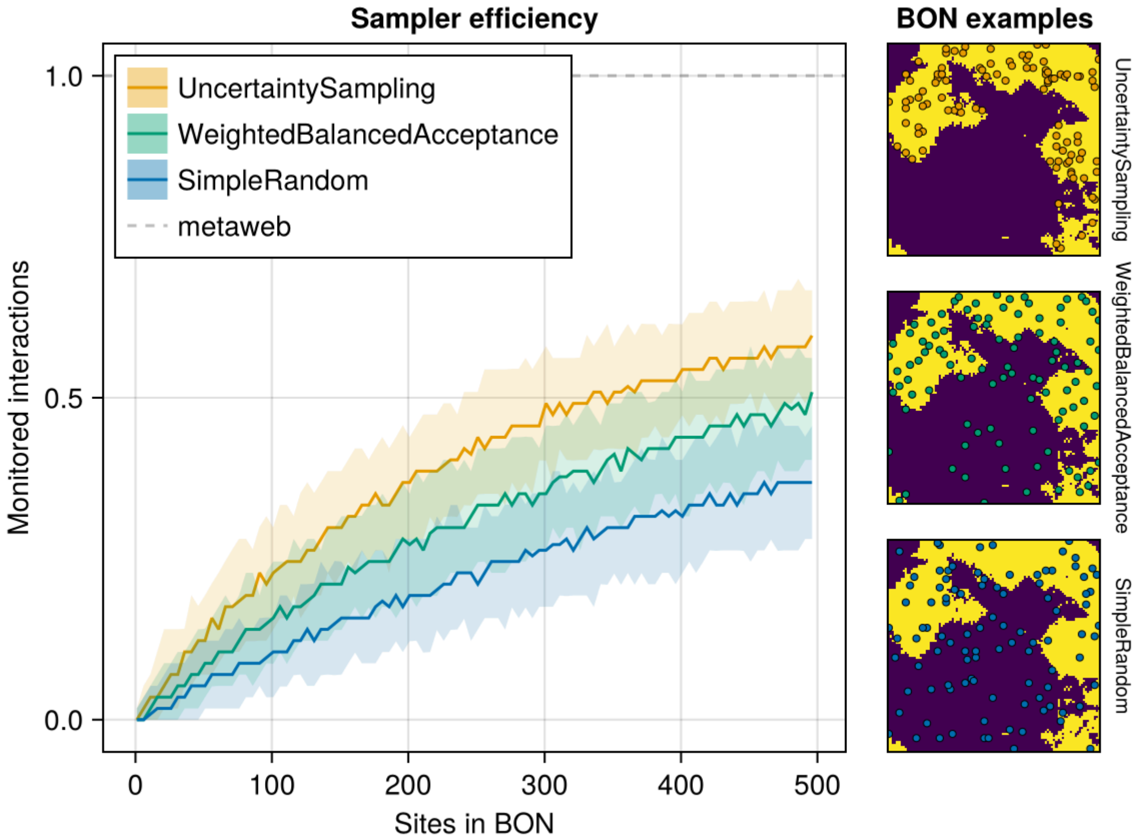

Figure 2: Efficiency of three sampler algorithms at sampling realized interactions for a given focal species within Biodiversity Observation Networks (BONs). The central line indicates indicates the median proportion for a given number of sites while the shaded bands delimit the 5th and 95th percentile across replicates. The right panels illustrate examples of BONs of 50 sites (coloured points) generated by each sampler over the focal species range (present at pixels in yellow).

Optimizing on the known species range, our proxy for an attainable level of information an interested user might have, was a more efficient strategy to retrieve the focal species' interactions than optimization based on species richness (or interaction richness), effectively sampling more than half of the species' interactions (Figure 3). In contrast, optimization based on the location of realized interactions, our best-case scenario requiring a higher level information, expectedly performed better, retrieving most of the species' interactions.

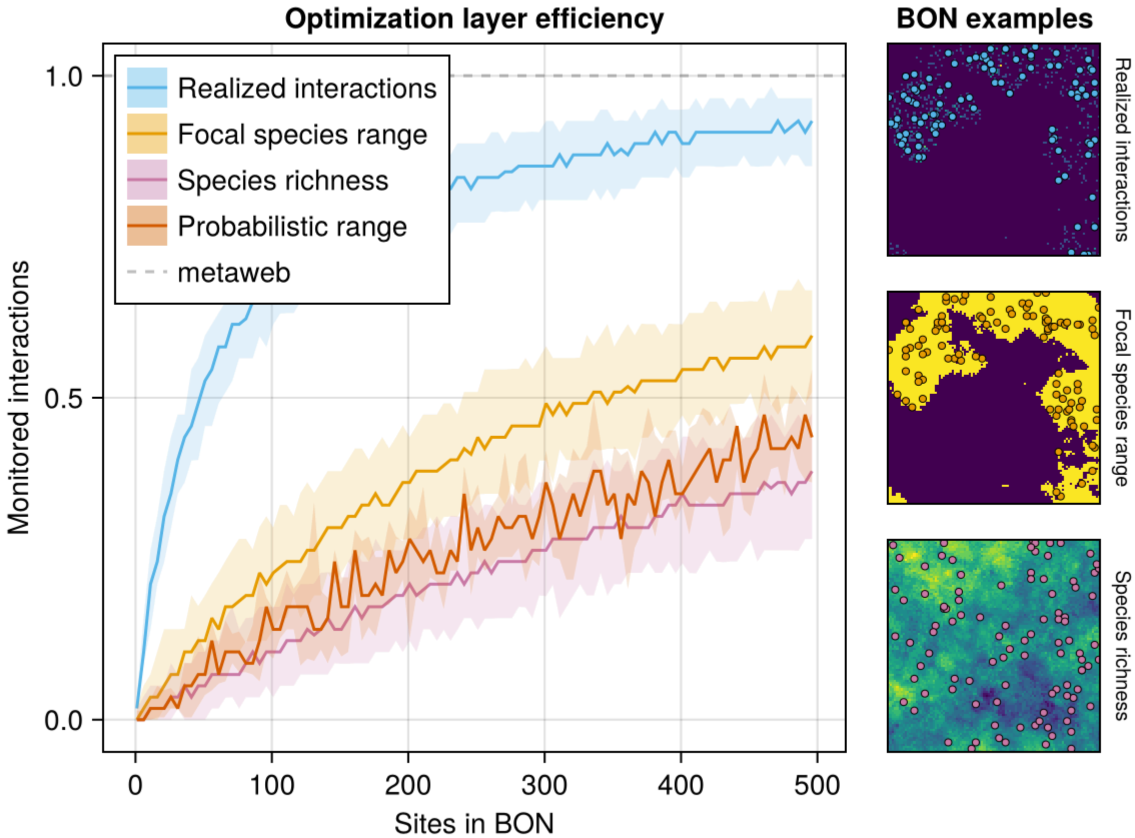

Figure 3: Efficiency of three optimization layers at sampling realized interactions for a given focal species within Biodiversity Observation Networks (BONs). The central line indicates indicates the median proportion for a given number of sites while the shaded bands delimit the 5th and 95th percentile across replicates. The right panels illustrate examples of BONs of 50 sites (coloured points) generated by each sampler over the focal species range (present at pixels in yellow).

Comparison of efficiency parameters across simulations

The distribution of efficiency parameters across independent simulations confirmed the interpretations based on Figure 2, Figure 3, but also highlighted a wide range of outcomes in efficiency between simulations. Among samplers, the median efficiency parameter indicated a better performance (lower parameter value) for Uncertainty Sampling, followed by Weighted Balanced Acceptance and Simple Random (Figure 4A). The actual efficiency values spanned a similar range across the possible sampler options with all simulations combined. Among optimization layers, optimizing on the focal species range performed better than on the probabilistic range, followed by species richness, although all three options spanned similar ranges with all simulations combined (Figure 4B). In contrast, optimizing based on the realized interactions distinctly and consistently performed better across all simulations.

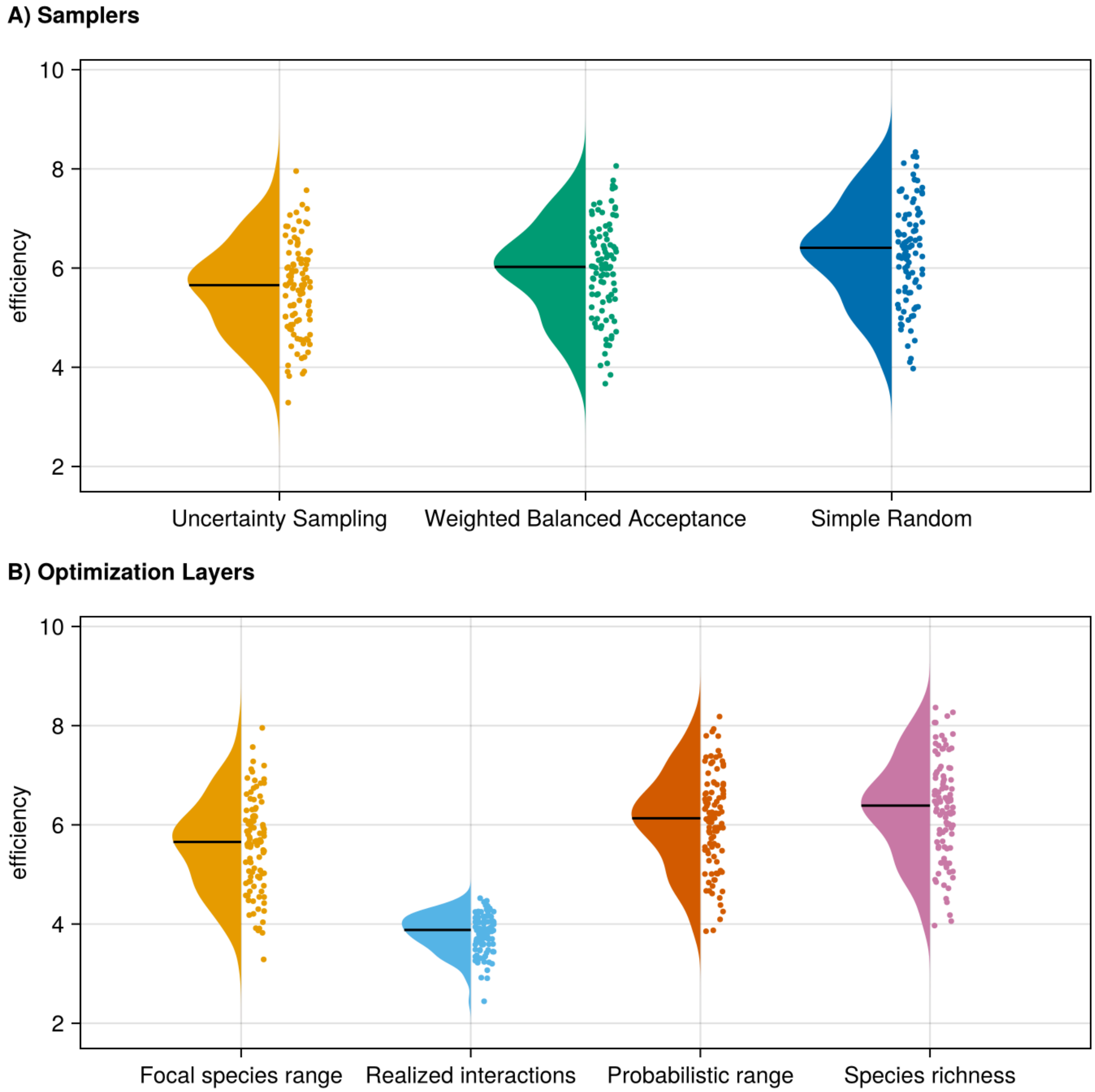

Figure 4: Distribution and density of efficiency parameters across 100 independent simulations for the three sampler options tested on the focal species range (A) and for the four optimization layer options using Uncertainty Sampling as the sampler (B). Uncertainty Sampling on panel A and Focal species range on panel B therefore represent the same data.

The fact that multiple options span similar ranges of efficiency values across independent simulations highlights the need to compare the efficiency within a given simulation. Comparing options one-by-one clearly highlighted a most efficient one in all case (Figure 5, Supp. Mat.), consistent with the showcased single-simulation example (Figure 2, Figure 3). Uncertainty Sampling was more efficient than Weighted Balanced Acceptance in 98 of 100 simulations, and both were more efficient than Simple Random in all simulations (Figure 5). Optimizing on the focal species range was always more efficient than on the probabilistic range or species richness (with probabilistic range more efficient than species richness in all but one simulation), but always less efficient optimizing on realized interactions.

Figure 5: Within-simulation comparison of efficiency between samplers and optimization layers. The comparison values is based on the difference of the definite integrals of the efficiency samplers curves described in Equation 3. Negative values (in blue) indicate that the compared option lead to a less efficient sampling than its reference, while positive values (in orange) indicate it was more efficient. References were chosen for narrative purposes and to simplify the number of comparisons displayed. All one-by-one comparisons are available in Supp. Mat.

While the simulations clearly highlight which sampler or optimization layer is most efficient across all simulations, efficiency parameters and differences vary over a wide range of values. We verified if this could be tied to the occupancy of the focal species in simulated landscape, measured as the number of cells where the species is present. We examined the relationship between occupancy and efficiency. Optimizing using Uncertainty Sampling on the focal species range and on realized interactions is especially efficient at low occupancy, as it targets the fewer cells where the focal species is present. As species occupancy increases, the difference between options decreases. When the focal species has high occupancy and is present almost everywhere across the landscape, Weighted Balanced Acceptance and Simple Random perform similarly to Uncertainty Sampling, as balancing sites across space or choosing at random becomes equivalent to targeting the species range. Optimizing based on species richness became more efficient as occupancy increased. A possible explanation is that at high occupancy, species range is not a restricting element and thus targeting sites with high richness directly leads to targeting sites with more interaction partners. On the contrary, at low occupancy, species range is an important restriction, and optimizing on richness of all species might target sites where the focal species is absent. Optimizing on the probabilistic species range behave similarly to species richness; at high occupancy, it is very likely the species will be present at the sites with high probabilities and therefore the sampler can more effectively filter out sites where the species is absent. At low occupancy, the focal species will be present in fewer locations with high probabilities, but might be absent from locations with intermediate probabilities, making it more challenging for the sampler to pick apart.

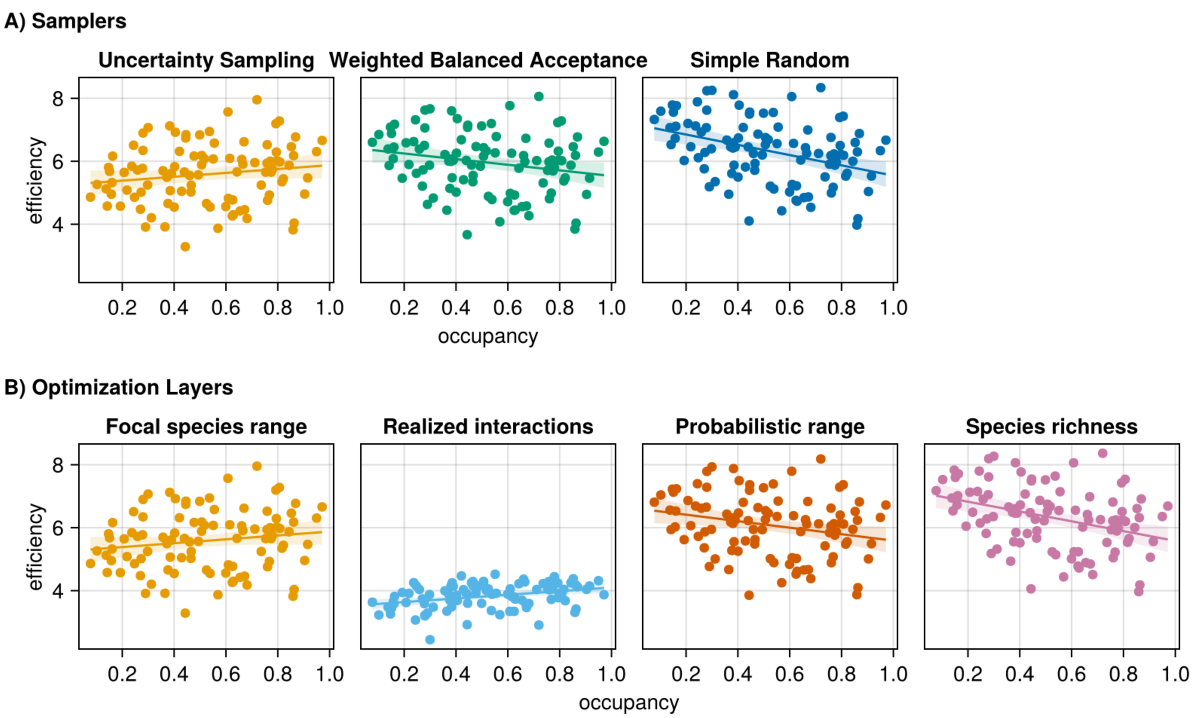

Figure 6: Relationship between the occupancy of the focal species and sampling efficiency for all optimization options. These results come from 100 independent simulations, and therefore 100 different sets of metawebs, species ranges, and occupancy of the focal species.

Discussion

What should be realistic expectations for sampling and monitoring of species interactions?

- Not as high as species richness and not as simple as co-occurrence of interacting species

- Expectations should be much lower given abundances, and our simulations highlight a way this could happen
- This means we can apply motoring to species interactions, with limits.

We can improve coverage with the right optimization method. But to do so, we need a clear and attainable sampling or monitoring objective.

- Again, we come back to a single species perspective as most attainable and understandable level of information while yielding relevant results.
- Optimizing on species range is an efficient strategy to retrieve its interactions (most attainable information), better than optimization based on species or interaction richness and surpassed only by knowledge of it's realized interactions

Our results offer insights into defining realistic expectations for interaction monitoring. With the right target and optimization strategy, available information, such as species ranges, can guide observation network designs to monitor species interactions with a reasonable number of sites.

References

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